



**SCHOOL OF COMPUTING AND ENGINEERING SCIENCES**  
**BACHELOR OF SCIENCE IN ELECTRICAL AND ELECTRONICS ENGINEERING**  
**BEE 5202: SUSTAINABLE ENERGY SYSTEMS**  
**IBERAFRICA INDUSTRIAL VISIT REPORT (CAT 2)**

**2026**

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Write a report **from your visit to IBERAFRICA Power Plant**. The report should include the sections highlighted in the format below.

**REPORT FORMAT:**

**1. Introduction**

Give a background of IBERAFRICA Power Plant, including its location, ownership, capacity, fuel type, technology used, etc.

**2. Schematic Diagram of the Power Plant**

Draw a schematic diagram of the power plant and explain the function of the various equipment in the plant.

**3. Transmission Substation**

Draw a layout diagram of the transmission substation and explain the function of the substation equipment.

**4. Protection, Switching, and Control Systems**

Identify the following components in the power plant (i.e. state whether they are present) and explain their functions. Also indicate their ratings and explain how they are connected to the power system.

- i). Distance relays
- ii). Differential relays
- iii). Over-current and over-voltage protection
- iv). Surge protection
- v). Lightning and harping equipment
- vi). SCADA system

## **5. Power Factor**

Identify the typical power factor values of the plant

Describe the methods used to correct for power factor at the plant

Explain the importance of power factor correction

## **6. Economics of power generation:**

### **6.1 Power Supply and Demand**

Discuss:

- i). The role of the power plant in meeting base load or peak load
- ii). Backup and dispatchable generation at the plant

### **6.2 Load Curves**

Describe daily and seasonal load curves and relate them to the operation of the plant

### **6.3 Cost of generation**

Discuss the cost components involved in electricity generation at the plant, including:

- i). Fuel cost contribution
- ii). Operation and maintenance costs
- iii). Capital costs
- iv). Cost per kWh of electricity generated

## **7. Policies and Regulations Affecting the Power Plant**

Discuss policies and regulations affecting the power plant, the regulatory bodies involved, and how these affect the operations of the plant.

## **8. Safety, Health, and Environmental Considerations**

Describe measures implemented in the power plant pertaining to:

- i). Occupational health and safety
- ii). Fire protection
- iii). Emission control
- iv). Environmental conservation
- v). Waste management
- vi). Etc

## **9. Efficiency of the Power Plant**

- i). Describe the overall efficiency of the plant
- ii). Identify factors that affect the efficiency of the plant
- iii). Propose how the efficiency of the plant can be improved

## **10. Sustainability**

Discuss the following:

- i). Environmental impact of the plant
- ii). Comparison with renewable alternatives
- iii). Potential integration with renewable systems
- iv). Role in Kenya's energy transition

## **11. Conclusion**

Reflect on the visit, highlighting lessons learnt.